Department of Electrical and Computer Engineering
Queen's University

ELEC-271 Digital Systems

Problem Set 4

1. Determine the decimal values of the following unsigned numbers:
   (a) $(0111011110)_2$ (b) $(3751)_8$ (c) $(A25F)_{16}$

2. Determine the decimal values of the following 2's complement numbers:
   (a) 1011100111 (b) 1111111110

3. Convert the decimal numbers 73, 1906, and -95 into signed 12-bit numbers in the following representations:
   (a) Sign and magnitude (b) 1's complement (c) 2's complement

4. Perform the following operations involving eight-bit 2's complement numbers and indicate whether arithmetic overflow occurs. Check your answers by converting to decimal sign-and-magnitude representation.

00110110	11011111	01110101	00110110
+01000101	+10111000	−11010110	−00101011

5. Prove that the circuit in Figure 5.5 of textbook implements the full-adder specified in Figure 5.4a.

6. Prove the validity of the expression Overflow = $c_n \oplus c_{n-1}$ for addition of n-bit signed numbers.

7. Determine the number of gates needed to implement an n-bit carry-lookahead adder, assuming no fan-in constraints. Use AND, OR, and XOR gates with any number of inputs.

8. Show the complete circuit for a four-bit hierarchical carry-lookahead adder, built using 2 two-bit blocks.

9. Write the VHDL code to describe the circuit block in Figure 5.32b of textbook.

10. Suppose that we want to determine how many of the bits in a three-bit unsigned number are equal to 1. Design the simplest circuit that can accomplish this task.

11. Repeat Q10 for a six-bit unsigned number.

12. Design the simplest circuit that determines whether 2 four-bit numbers are equal.